Netflix! What started in 1997 as a DVD rental service has since exploded into one of the largest entertainment and media companies.

Given the large number of movies and series available on the platform, it is a perfect opportunity to flex your exploratory data analysis skills and dive into the entertainment industry.

You work for a production company that specializes in nostalgic styles. You want to do some research on movies released in the 1990's. You'll delve into Netflix data and perform exploratory data analysis to better understand this awesome movie decade!

You have been supplied with the dataset `netflix_data.csv`, along with the following table detailing the column names and descriptions. Feel free to experiment further after submitting!

The data

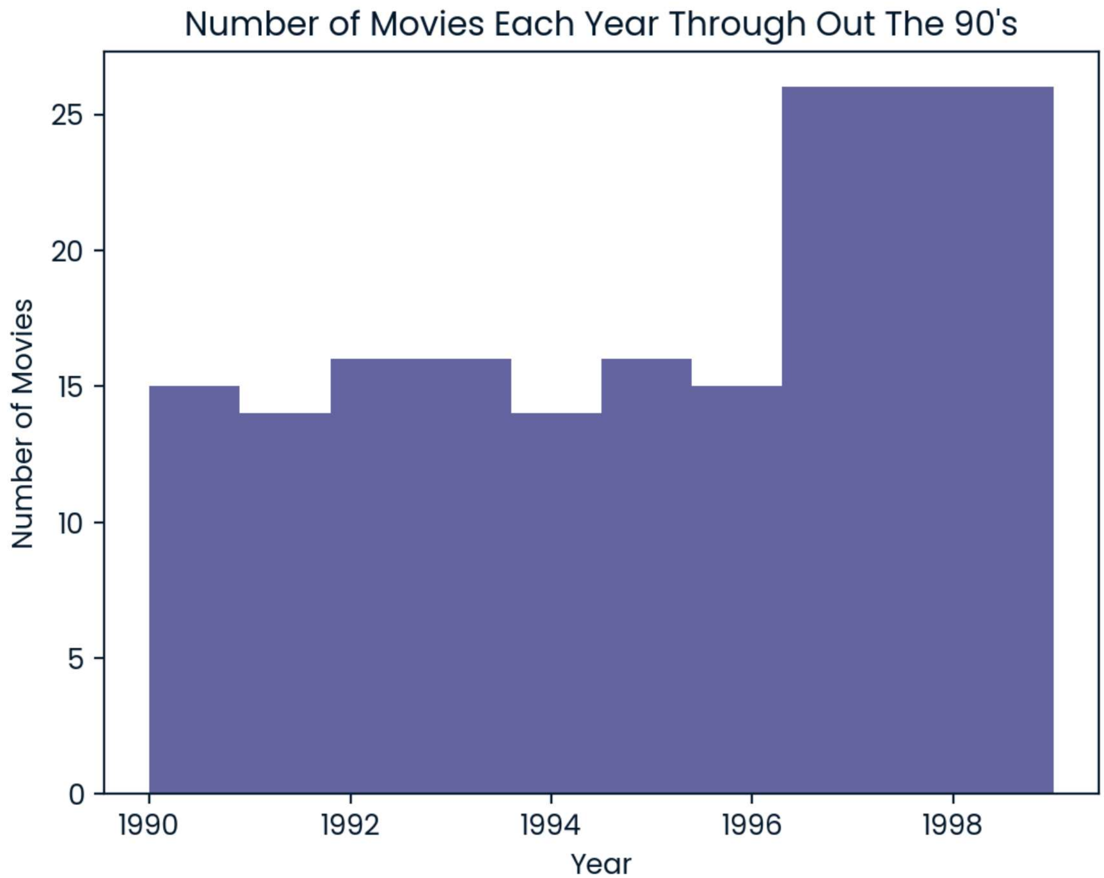
netflix_data.csv

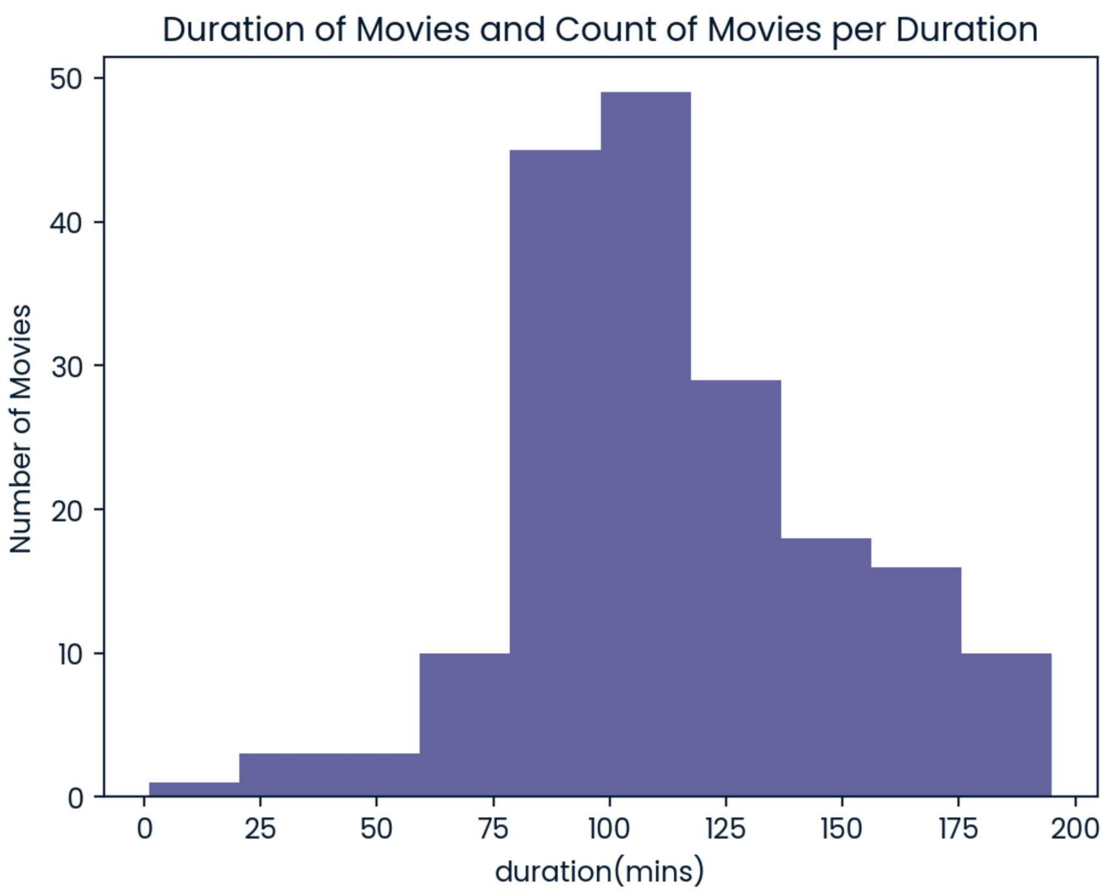
Column	Description
`show_id`	The ID of the show
`type`	Type of show
`title`	Title of the show
`director`	Director of the show
`cast`	Cast of the show
`country`	Country of origin
`date_added`	Date added to Netflix
`release_year`	Year of Netflix release
`duration`	Duration of the show in minutes
`description`	Description of the show
`genre`	Show genre

```
#Importing pandas and plt
import pandas as pd
import matplotlib.pyplot as plt

# Add netflix_data.csv
netflix_df = pd.read_csv("netflix_data.csv")
print(netflix_df)

# Checking for duplicates.
print("Number of duplicates:", netflix_df.duplicated().sum())

# Filtering the data for movies released in the 1990's
netflix_subset = netflix_df[netflix_df["release_year"].between(1990, 1999)]

# If you want to overwrite netflix_df with the filtered data, do:
netflix_df = netflix_subset
#Data vizulization of the number of movies in each year through the 90's
plt.hist(netflix_df["release_year"])
plt.xlabel("Year")
plt.ylabel("Number of Movies")
plt.title("Number of Movies Each Year Through Out The 90's")
plt.show()
#Netflix featured more movies in the late 90's compared to the early 90's

#Question 1 What was the most frequent movie duration in the 1990s? Save an approximate answer as an integer called duration (use 1990 as the decade's start year).
duration = int(netflix_df["duration"].mode()[0])
plt.hist(netflix_df["duration"])
plt.xlabel("duration(mins)")
plt.ylabel("Number of Movies")
plt.title("Duration of Movies and Count of Movies per Duration")
plt.show()
netflix_df["duration"].value_counts()
netflix_df["duration"].mode().sum()
print(netflix_df["duration"].mode().sum())
#The most frequent duration is 94 mins


#Question 2 A movie is considered short if it is less than 90 minutes. Count the number of short action movies released in the 1990s and save this integer as short_movie_count.

#Creating a bar chart for genres
genre_counts = netflix_df["genre"].value_counts()

plt.barh(genre_counts.index, genre_counts.values)
```

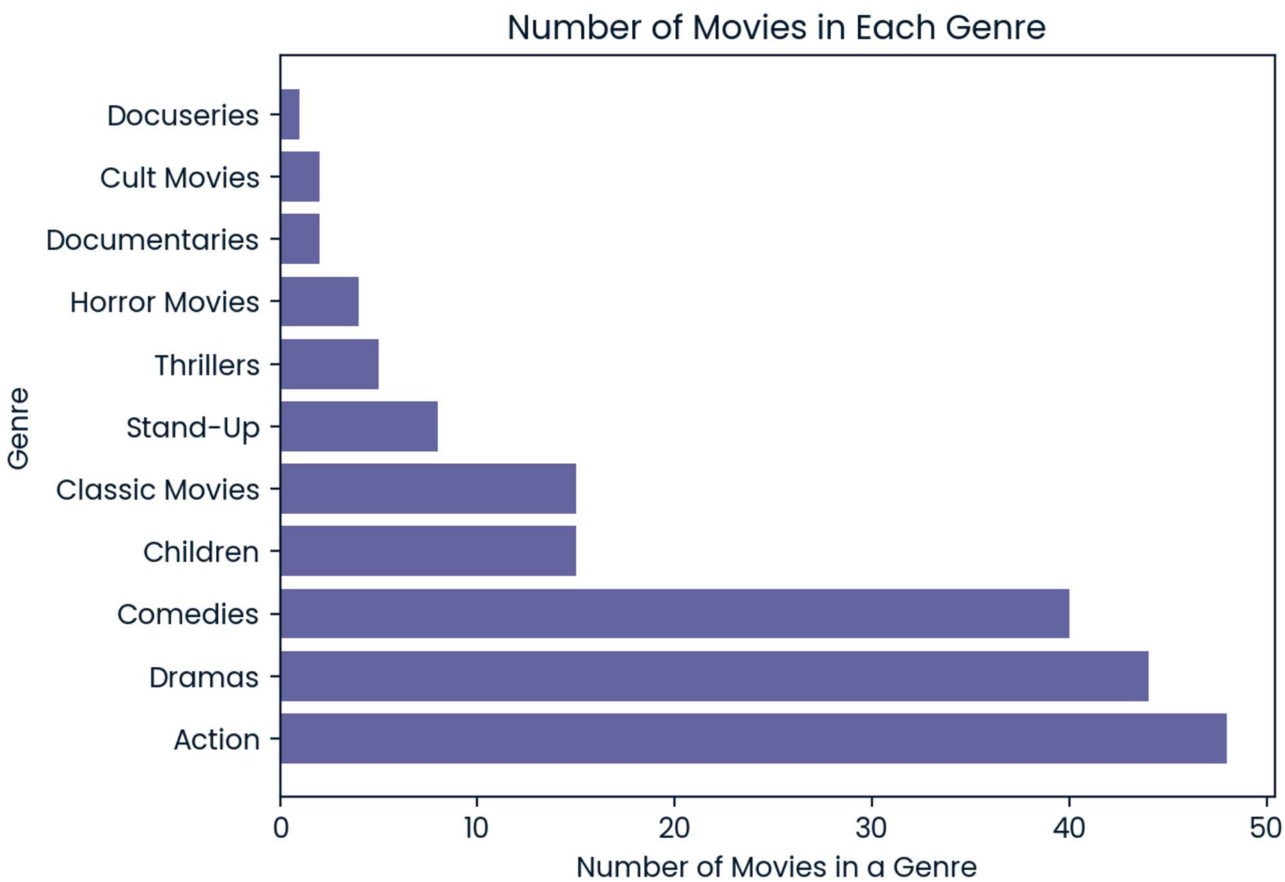
```python
plt.xlabel("Number of Movies in a Genre")
plt.ylabel("Genre")
plt.title("Number of Movies in Each Genre")
plt.show

#Action movies have the highest amount of Movies any genre.

#Subsetting the short actions movies with less than 90 mins
short_action_movies= (netflix_df["genre"] == "Action")& (netflix_df["duration"] <
90)
short_action_movies
short_movie_count = short_action_movies.sum()
print(short_movie_count)
#Only 7 action movies were shorter than 90 mins in the 1990's, therefore it does not
seem that action movies are going to get shorter in the foreseable future
```

```
     show_id  ...             genre
0         s2  ...            Dramas
1         s3  ...     Horror Movies
2         s4  ...            Action
3         s5  ...            Dramas
4         s6  ...  International TV
...      ...  ...               ...
4807   s7779  ...          Comedies
4808   s7781  ...            Dramas
4809   s7782  ...          Children
4810   s7783  ...            Dramas
4811   s7784  ...            Dramas

[4812 rows x 11 columns]
Number of duplicates: 0
```


Number of Movies Each Year Through Out The 90's
Number of Movies
0
5
10
15
20
25
1990
1992
1994
1996
1998
Year


Duration of Movies and Count of Movies per Duration
Number of Movies
0
10
20
30
40
50
0
25
50
75
100
125
150
175
200
duration(mins)

94
7


Number of Movies in Each Genre
Docuseries
Cult Movies
Documentaries
Horror Movies
Thrillers
Stand-Up
Classic Movies
Children
Comedies
Dramas
Action
Genre
0
10
20
30
40
50
Number of Movies in a Genre